

1. Create an `ArrayList<String>` of student names

Add at least 5 names

Print all names

Remove the 3rd name

2. Sum of Integers

Create a `List<Integer>`

Add 10 numbers

Calculate the sum and average

3. Reverse a list:

Given a list of integers, reverse it manually

4. Unique Words Counter

Given a sentence, split it into words

Store them in a `Set<String>`

Print unique words and their count

5. Word Frequency Counter

Input: "apple banana apple orange banana apple"

Use a Map<String, Integer>

Count how many times each word appears

6. Phone Book

Create a Map<String, String> (name → phone)

Add entries

Search by name

Print all entries

7. Students management system

-given the following class:

```
class Student {
```

```
    String name;
```

```
    int grade;
```

```
}
```

Store students in a List<Student>

Print all students

Find the student with the highest grade

8.Sort students by:

Name (alphabetically)

Grade (descending)

9.Remove Duplicates from Objects

Given a list of students (with duplicates), remove duplicates

Optional:

10. LRU Cache (Simplified)

Implement a basic cache using LinkedHashMap

11. Merge Two Maps

Merge two `Map<String, Integer>`

If keys overlap → sum values